SET 30: MOBILE APP USAGE ANALYSIS

Dataset: Mobile_App_Usage.csv

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	4	2025-03-14
U02	Female	22	6.0	25	3.8	3	2025-01-08
U03	Male	19	3.2	12	1.6	5	2025-01-08
U04	Female	21	7.1	30	4.5	3	2025-02-11
U05	Male	23	2.8	10	1.2	5	2025-02-11
U06	Female	20	5.4	22	3.1	2	2025-03-14

PROGRAM 1: Histogram and Density Plot of Screen Time

Question: Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

```
# Import the dataset
usage <- read.csv("Mobile_App_Usage.csv")
usage$Usage_Date <- as.Date(usage$Usage_Date)
str(usage)

# Histogram with density curve
hist(usage$Screen_Time, freq = FALSE, breaks = seq(2, 8, by = 1),
     main = "Distribution of Screen Time",
     xlab = "Screen Time (hrs)", ylab = "Density",
     col = "#A0CBE8", border = "white", ylim = c(0, 0.4))
lines(density(usage$Screen_Time), col = "#E15759", lwd = 2)
abline(v = mean(usage$Screen_Time), lty = 2, lwd = 2, col = "#4E79A7")
legend("topright", legend = c("Density curve", "Mean"),
       col = c("#E15759", "#4E79A7"), lty = c(1, 2), lwd = 2)

# Summary statistics
summary(usage$Screen_Time)
sd(usage$Screen_Time)
```

Output 1: Histogram and Density Plot of Screen Time



Observation: Screen time ranges from 2.8 to 7.1 hours with a mean of 4.83, a median of 4.95 and a standard deviation of 1.66. The histogram is spread broadly with no single sharp peak; the tallest bar is 5 to 6 hours, where two users fall, and the density curve is flat and wide, peaking near 5 hours. The mean sits just below the median, so the distribution is close to symmetric with no extreme outliers. There is a visible split between lighter users (2.8 to 4.5 hours) and heavier users (5.4 to 7.1 hours). With six users the shape is only indicative.

PROGRAM 2: Scatter Plot of Data Used vs Screen Time with Correlation

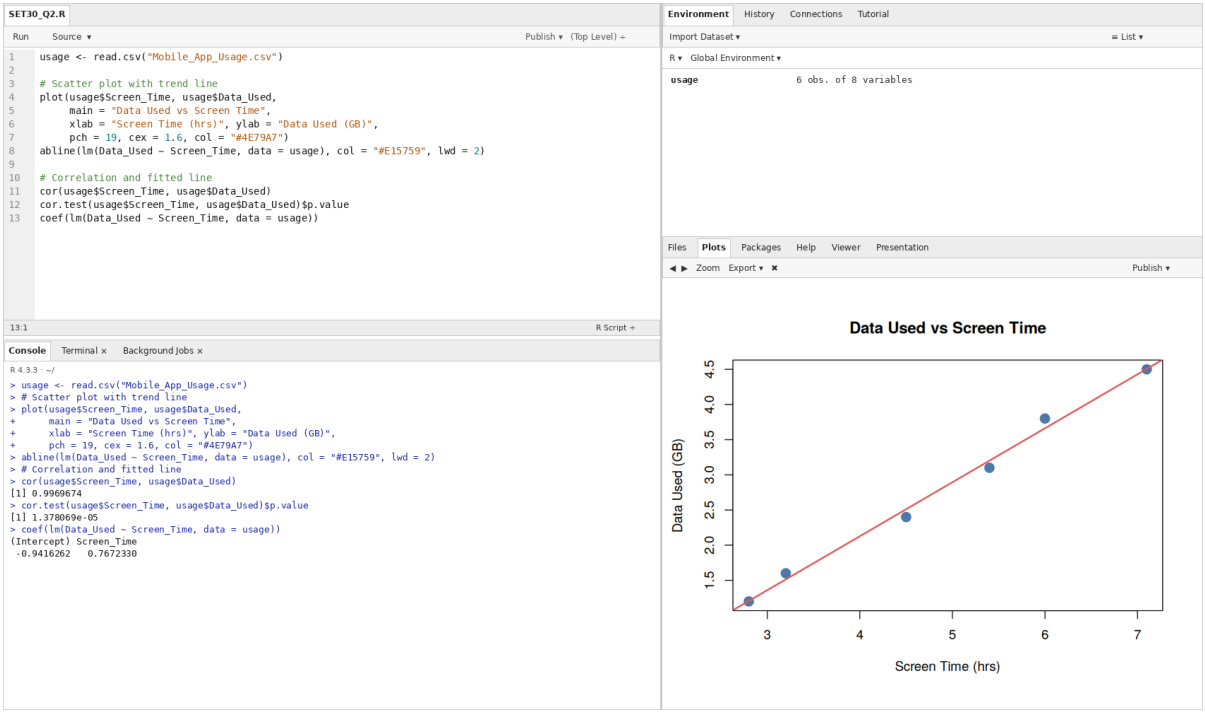
Question: Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

```
usage <- read.csv("Mobile_App_Usage.csv")

# Scatter plot with trend line
plot(usage$Screen_Time, usage$Data_Used,
     main = "Data Used vs Screen Time",
     xlab = "Screen Time (hrs)", ylab = "Data Used (GB)",
     pch = 19, cex = 1.6, col = "#4E79A7")
abline(lm(Data_Used ~ Screen_Time, data = usage), col = "#E15759", lwd = 2)

# Correlation and fitted line
cor(usage$Screen_Time, usage$Data_Used)
cor.test(usage$Screen_Time, usage$Data_Used)$p.value
coef(lm(Data_Used ~ Screen_Time, data = usage))
```

Output 2: Scatter Plot with Trend Line



Observation: The points lie almost exactly on the trend line, so Data_Used and Screen_Time have a very strong positive linear relationship (r = 0.997, p = 0.00001). The fitted line is Data_Used = -0.94 + 0.767 x Screen_Time, which means each extra hour on screen is associated with about 0.77 GB more data. This is expected, since longer sessions stream and load more content. The relationship is based on six users, and a strong correlation does not by itself prove that screen time causes the extra data use.

PROGRAM 3: Average Satisfaction by Gender

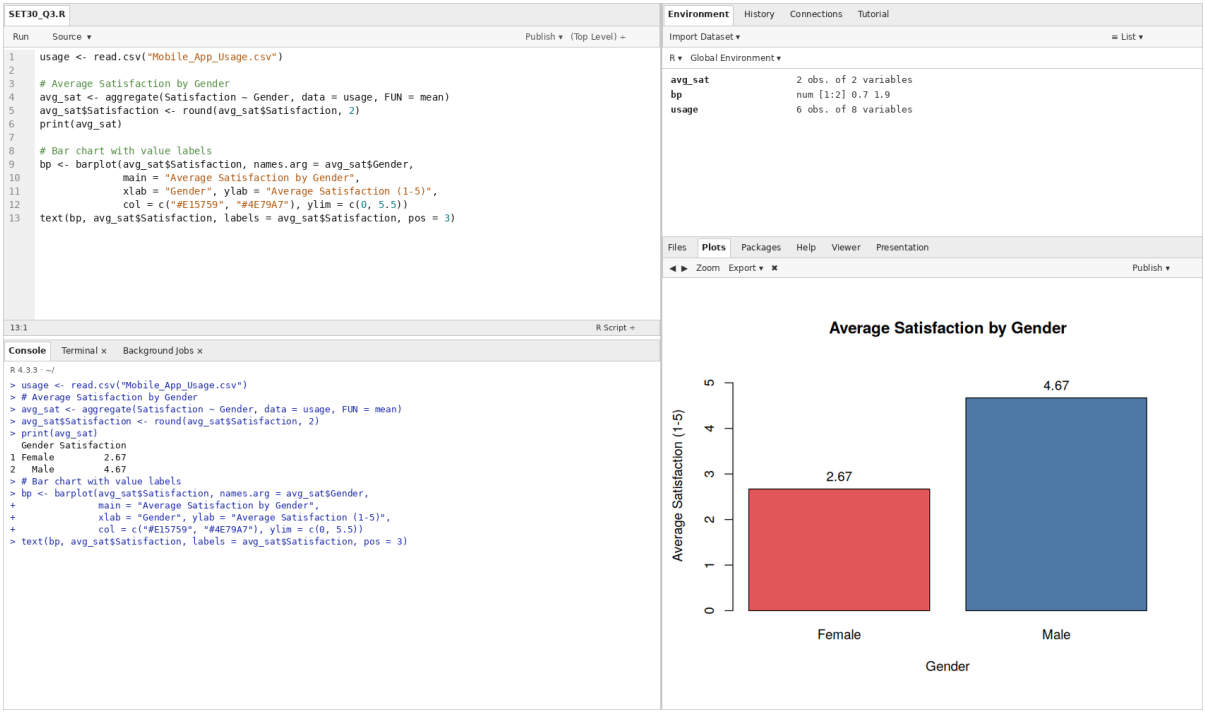
Question: Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

```
usage <- read.csv("Mobile_App_Usage.csv")

# Average Satisfaction by Gender
avg_sat <- aggregate(Satisfaction ~ Gender, data = usage, FUN = mean)
avg_sat$Satisfaction <- round(avg_sat$Satisfaction, 2)
print(avg_sat)

# Bar chart with value labels
bp <- barplot(avg_sat$Satisfaction, names.arg = avg_sat$Gender,
              main = "Average Satisfaction by Gender",
              xlab = "Gender", ylab = "Average Satisfaction (1-5)",
              col = c("#E15759", "#4E79A7"), ylim = c(0, 5.5))
text(bp, avg_sat$Satisfaction, labels = avg_sat$Satisfaction, pos = 3)
```

Output 3: Average Satisfaction by Gender



Observation: Male users report an average satisfaction of 4.67 against 2.67 for female users, a gap of 2 points on the 1 to 5 scale. The female users in this sample also have heavier screen time (average 6.17 hours against 3.5 hours for the male users) and higher data use, so lower satisfaction goes together with heavier app use. This could point to problems such as battery drain, data cost or app performance affecting heavy users, and it would be worth investigating. With three users per gender, this is a pattern in the sample rather than a firm conclusion about all users.

PROGRAM 4: Interactive Mobile App Usage Dashboard in Tableau

Question: Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.

Tableau Steps

1. Open Tableau, choose Connect > Text file and select Mobile_App_Usage.csv.

2. On the Data Source tab, set Usage_Date to Date, Gender to String, and the other fields to numbers.

3. Create a worksheet named "Bubble Chart". Drag Screen_Time to Columns and Data_Used to Rows, then drag User_ID to Detail so each user is one mark.

4. Choose Circle in the Marks card. Drag Gender to Color and App_Usage_Count to Size. Add User_ID to Label and add all fields to the Tooltip. Title it "Screen Time vs Data Used".

5. Create a worksheet named "Usage Trend". Drag Usage_Date to Columns (Exact Date, continuous) and App_Usage_Count to Rows (Average). Choose Line in the Marks card and show mark labels.

6. Optionally add Screen_Time as a second measure (Dual Axis) and drag Gender to Color to compare trends. Title it "Usage Trend by Date".

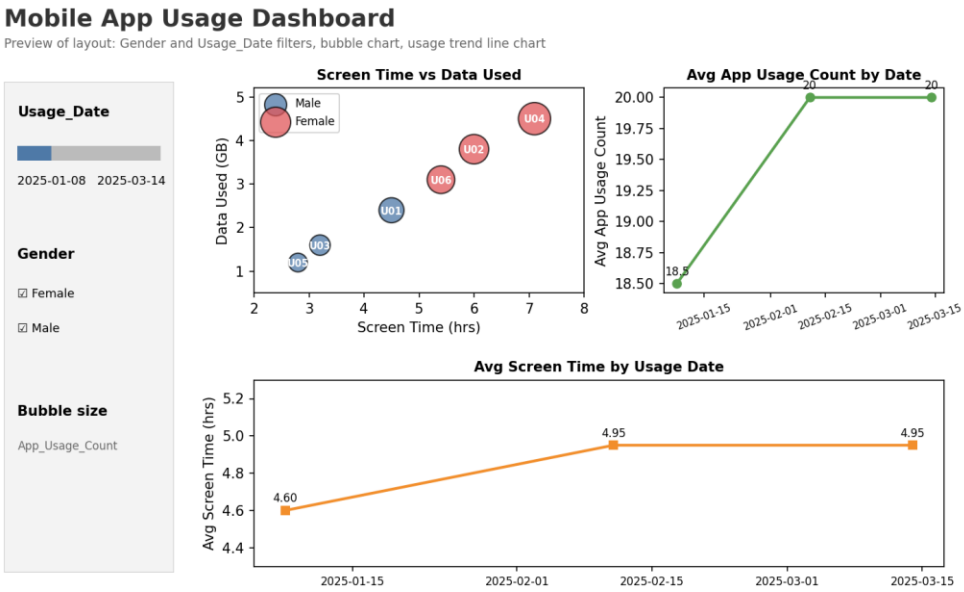
7. Add the filters: drag Gender and Usage_Date to Filters (Usage_Date as Range of Dates). Right-click each, choose Show Filter, and set Apply to Worksheets > All Using This Data Source.

8. Click New Dashboard, set the size to Automatic, and drag in the bubble chart and the line chart. Add the title "Mobile App Usage Dashboard".

9. Select the bubble chart and enable Use as Filter, and show the Gender and Usage_Date filters on the dashboard.

10. Test the dashboard by switching Gender, changing the date range, and clicking a bubble to see the line chart update.

Output 4: Mobile App Usage Dashboard (layout preview)



Observation: The bubble chart shows the female users (red) as the larger bubbles at the upper right (high screen time, high data use, high app usage) and the male users (blue) as the smaller bubbles at the lower left. The trend line chart shows average app usage rising from 18.5 in January to 20 in February and staying at 20 in March, while average screen time rises from 4.6 to 4.95 hours and then stays level, so usage grew early in the period and then levelled off. The Gender filter and the Usage_Date range filter allow the viewer to compare the two groups and focus on any month. Each date has only two users, so the trend is only indicative.

Procedure

1. Import Mobile_App_Usage.csv into R using read.csv() and convert Usage_Date with as.Date().

2. Plot the histogram of Screen_Time with hist() and add the density curve with lines(density()).

3. Draw the scatter plot of Data_Used against Screen_Time, compute cor() and add the trend line using lm() and abline().

4. Calculate the average Satisfaction by Gender with aggregate() and draw the bar chart with barplot() and value labels using text().

5. Import the dataset into Tableau and build the bubble chart and usage trend line chart.

6. Add the Gender and Usage_Date filters, combine the sheets into a dashboard and test the interactivity.

Result

The Mobile App Usage data was successfully analysed in R and Tableau. Screen time is broadly spread (mean 4.83 hours), data used rises almost linearly with screen time ($r = 0.997$), male users report higher average satisfaction (4.67) than female users (2.67), and an interactive Tableau dashboard with a bubble chart, usage trend line and Gender and Usage_Date filters was designed.